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EMBEDDED ELECTROSTATIC CHUCK (ESC)

Abstract

Embodiments disclosed herein may include an apparatus that includes a body with a first surface and a second surface opposite from the first surface. In an embodiment, the body is an electrically conductive material. A hole may be formed into the first surface of the body, and a ceramic plate may be on the second surface of the body. An electrically conductive electrode may be embedded in the ceramic plate. In an embodiment, the apparatus may further include a pin that is electrically conductive and positioned in the hole. The pin may be electrically isolated from the body, and the pin may be electrically coupled to the electrode embedded in the ceramic plate.

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Background/Summary

BACKGROUND

1) Field

[0001] Embodiments relate to the field of semiconductor manufacturing and, in particular, to electrostatic chucks (ESCs) with an RF feed that is electrically isolated from a DC voltage supply.

2) Description of Related Art

[0002] In semiconductor manufacturing processes, the wafer (e.g., a silicon wafer) or other substrate is coupled to a pedestal during various processes. For example, the wafer may be secured to the pedestal during plasma dicing processes, etching processes, deposition processes, and/or the like. In some embodiments, the wafer is secured to the pedestal through the use of an electrostatic chuck (ESC). The ESC provides an electrostatic force that attracts the wafer to the chuck in order to prevent movement of the wafer during processing.

[0003] Typically, the ESC is fed several inputs from an underlying cathode assembly. In existing solutions, an RF feed (for applying an RF bias) and a DC supply (for generating a chucking force) are provided to the ESC through a single input. The coupling of the RF feed and the DC supply can lead to high current leakage. For example, electrical current from the DC supply can leak into the plasma environment.

SUMMARY

[0004] Embodiments disclosed herein may include an apparatus that includes a body with a first surface and a second surface opposite from the first surface. In an embodiment, the body is an electrically conductive material. A hole may be formed into the first surface of the body, and a ceramic plate may be on the second surface of the body. An electrically conductive electrode may be embedded in the ceramic plate. In an embodiment, the apparatus may further include a pin that is electrically conductive and positioned in the hole. The pin may be electrically isolated from the body, and the pin may be electrically coupled to the electrode embedded in the ceramic plate.

[0005] Embodiments disclosed herein may include a method of placing a substrate on an electrostatic chuck (ESC). In an embodiment, the ESC has a DC input that is electrically isolated from an RF input. In an embodiment, the method include applying a chucking force to the substrate by activating the DC input, and processing the substrate in a plasma environment during application of an RF bias applied by the RF input. In an embodiment, the method may include releasing the chucking force, and removing the substrate from the ESC.

[0006] Embodiments disclosed herein may include a semiconductor processing tool that includes a chamber suitable for maintaining a vacuum environment within the chamber and an electrostatic chuck (ESC) within the chamber. The ESC may include a body that is electrically conductive, and a ceramic plate on the body. An electrically conductive electrode may be embedded within the ceramic plate. In an embodiment, a DC input is electrically coupled to the electrode, and the DC input is electrically isolated from the body. The ESC may further include an RF input that is electrically coupled to the body.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1**A is a perspective view illustration of an electrostatic chuck (ESC) that comprises an metallic body with a ceramic plate with an embedded electrode, in accordance with an embodiment.

[0008] FIG. **1**B is a cross-sectional illustration of an ESC with a single input for the RF feed and the DC input, in accordance with an embodiment.

[0009] FIG. **2**A is a cross-sectional illustration of an ESC with a DC input and an RF feed that are electrically isolated from each other, in accordance with an embodiment.

[0010] FIG. **2**B is a cross-sectional illustration of an ESC with an underlying facility plate for

coupling the DC input and the RF feed to the ESC, in accordance with an embodiment.
[0011] FIG. **3**A is a cross-sectional illustration of a portion of an ESC with openings to accommodate a DC input and an RF feed, in accordance with an embodiment.
[0012] FIG. **3**B is a zoomed in cross-sectional illustration of the DC input that is electrically isolated from the metallic body of the ESC, in accordance with an embodiment.
[0013] FIG. **3**C is a cross-sectional illustration of the ESC with the DC input and the RF feed coupled to the ESC, in accordance with an embodiment.
[0014] FIG. **3**D is a cross-sectional illustration of the ESC with an underlying facility plate for coupling the DC input and the RF feed to the ESC, in accordance with an embodiment.
[0015] FIG. **4** is a cross-sectional illustration of a semiconductor processing tool that comprises an ESC with a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.
[0016] FIG. **5** is a process flow diagram of a method for processing a substrate with an ESC that comprises a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.
[0017] FIG. **6** is a process flow diagram of a method for assembling the ESC to a pedestal with a blind assembly process, in accordance with an embodiment.
[0018] FIG. **7** illustrates a block diagram of an exemplary computer system that may be used in conjunction with a processing tool, in accordance with an embodiment.

DETAILED DESCRIPTION

[0019] Embodiments described herein include apparatuses and methods for using an electrostatic chuck (ESC) with an RF feed that is electrically isolated from a DC voltage supply. In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments. It will be apparent to one skilled in the art that embodiments may be practiced without these specific details. In other instances, well-known aspects are not described in detail in order to not unnecessarily obscure embodiments. Furthermore, it is to be understood that the various embodiments shown in the accompanying drawings are illustrative representations and are not necessarily drawn to scale.
[0020] Various embodiments or aspects of the disclosure are described herein. In some implementations, the different embodiments are practiced separately. However, embodiments are not limited to embodiments being practiced in isolation. For example, two or more different embodiments can be combined together in order to be practiced as a single device, process, structure, or the like. The entirety of various embodiments can be combined together in some instances. In other instances, portions of a first embodiment can be combined with portions of one or more different embodiments. For example, a portion of a first embodiment can be combined with a portion of a second embodiment, or a portion of a first embodiment can be combined with a portion of a second embodiment and a portion of a third embodiment.
[0021] The embodiments illustrated and discussed in relation to the figures included herein are provided for the purpose of explaining some of the basic principles of the disclosure. However, the scope of this disclosure covers all related, potential, and/or possible, embodiments, even those differing from the idealized and/or illustrative examples presented. This disclosure covers even those embodiments which incorporate and/or utilize modern, future, and/or as of the time of this writing unknown, components, devices, systems, etc., as replacements for the functionally equivalent, analogous, and/or similar, components, devices, systems, etc., used in the embodiments illustrated and/or discussed herein for the purpose of explanation, illustration, and example.
[0022] As noted above, existing electrostatic chuck (ESC) structures are susceptible to significant current leakage. The leakage may result from the current (e.g., DC or RF) leaking away from the ESC and coupling with the plasma within a chamber and/or otherwise escaping through other electrical pathways. One problem with leakage is that chucking force is difficult to control. In some instances, current leakage during a processing operation within the chamber may result in

dechucking of the wafer. That is, the chucking force dips below a given threshold, and the wafer becomes free to move, bend, or otherwise displace relative to the surface of the ESC. This movement is detrimental because the processing conditions are highly tuned, and the movement may result in non-uniform treatment of the wafer or damage to the wafer or ESC.

[0023] To prevent dechucking, the wafer is often “overchucked” so that the chucking force is increased beyond what would otherwise be necessary. This can lead to damage to the wafer and/or damage to the ESC itself. For example, an overchucked wafer can be cracked, chipped, deformed, or otherwise damaged. Overchucking may also lead to excessive wear on the ESC. This may require more frequent repair, replacement, and/or refurbishing of the ESC. As such, the leakage current can lead to increases in cost of ownership of a semiconductor processing tool, and/or an increase in manufacturing costs due to damaged wafers. The presence of leakage current also reduces the efficiency of the system. That is, more energy is needed in order to run a given process when overchucking is necessary to account for leakage. This also increases costs and can generate environmental impact issues.

[0024] Referring now to FIG. **1**A, a perspective view illustration of an ESC **100** is shown, in accordance with an embodiment. In an embodiment, the ESC **100** may comprise a body **101**. The body **101** may be an electrically conductive material. For example, the body **101** may comprise aluminum, or the like. The body **101** may have a first surface **103** (e.g., a bottom surface) and a second surface **104** (e.g., a top surface) opposite from the bottom surface. In an embodiment, the body has a cylindrical shape that is suitable for supporting a wafer, such as a standard silicon wafer. For example, a diameter of the body **101** may be at least 200 mm or larger, at least 300 mm or larger, at least 450 mm or larger, or at least 750 mm or larger.

[0025] In an embodiment, a ceramic plate **105** may be provided on the second surface **104** of the body **101**. The ceramic plate **105** may be set into a recess of the second surface **104** so that the top surface of the ceramic plate **105** is substantially coplanar with a top surface of the body **101**. Though, in other embodiments, the ceramic plate **105** may have a top surface above or below the top surface of the body **101**.

[0026] In an embodiment, the ceramic plate **105** may comprise an electrically conductive electrode (not visible in FIG. **1**A). The electrode may be embedded within a thickness of the ceramic plate **105**. The electrode is coupled to a DC input in order to generate a chucking force that attracts and secures a wafer (not shown) to the ceramic plate **105**. In an embodiment, the ceramic plate may comprise any suitable material, such as aluminum nitride, aluminum oxide, or the like.

[0027] Referring now to FIG. **1**B, a cross-sectional illustration of the ESC **100** in FIG. **1**A is shown, in accordance with an embodiment. As shown, the ceramic plate **105** is set into a recess along the second surface **104** of the body **101**. The ceramic plate **105** may also comprise an electrode **108**. The electrode **108** may be an electrically conductive material, such as copper or the like. The electrode **108** may be a conductive plate, a conductive mesh, or have any other conductive pattern that is distributed though the ceramic plate **105**. In the illustrated embodiment, the electrode **108** is set at a midpoint along a thickness direction of the ceramic plate **105**. Though, in other embodiments the electrode **108** may be closer to a top of the ceramic plate **105** or closer to a bottom of the ceramic plate **105**. When the electrode **108** is embedded within the ceramic plate **105** as shown in FIG. **1**B, the ESC **100** may sometimes be referred to as an embedded ESC **100**.

[0028] In an embodiment, the electrode **108** may be electrically coupled to an input **112**. The input **112** may be an electrically conductive pin that passes through at least a thickness of the body **101**. The input **112** may pass through a portion of the ceramic plate **105** in order to directly contact the electrode **108**. In other embodiments, the input **112** may be electrically coupled to the electrode **108** by a via (not shown) that passes through at least a portion of the thickness of the ceramic plate **105**.

[0029] In an embodiment, the input **112** may provide both the DC input and the RF feed to the ESC. For example, DC current supplied to the input **112** may be provided to the electrode **108** in order to generate a chucking force on a wafer (not shown) that is supported by the ESC **100**. The input **112** may also receive an RF signal that is transferred to the electrically conductive body **101** of the ESC **100**. For example, the body **101** directly contacts the input **112** as well. Since the DC input and the RF feed are not electrically isolated from each other, leakage is more prevalent in the ESC **100** compared to embodiments that will be described in greater detail herein. This may lead to issues with dechucking, excessive energy consumption, and/or the like.

[0030] In an embodiment, the ESC **100** may also comprise fluidic channels **115**. The fluidic channels **115** may be provided at a bottom of the ESC **100** and sealed with a lid **116**. The fluidic channels **115** may be suitable for flowing gas and/or liquid within the body **101**. This can be used for cooling or other thermal control of the ESC **100**. In some embodiments, multiple zones of fluidic channels **115** may be used in order to provide more refined thermal control across an entire surface of the ESC **100**.

[0031] As described above, the ESC **100** in FIGS. **1**A and **1**B suffers from current leakage due to the combined input **112** for the DC current and RF current. Accordingly, embodiments disclosed herein may include an ESC that has a DC input line that is separate from the RF feed. More particularly, the DC voltage applied to the electrode within the ceramic plate can be electrically isolated from the conductive body of the ESC. Similarly, the RF feed can be provided directly to the conductive body of the ESC without being superimposed on the DC input line. Due to the electrical isolation, the leakage current can be reduced.

[0032] An example of such an ESC **200** is shown in FIG. **2**A. FIG. **2**A is a cross-sectional illustration of an ESC **200** that comprises an electrically conductive body **201**. The electrically conductive body **201** may be aluminum or another suitable metallic material. The conductive body **201** may be cylindrical for supporting wafers in a semiconductor processing environment. For example, the conductive body **201** may have a diameter that is larger than standard wafer form factors (e.g., larger than 200 mm, 300 mm, 450 mm, or the like). In an embodiment, the conductive body **201** may have a first surface **203** (e.g., a bottom surface) and a second surface **204** (e.g., a top surface) that is opposite from the first surface **203**.

[0033] In an embodiment, the ESC **200** may comprise fluidic channels **215**. The fluidic channels **215** may be provided at a bottom of the ESC **200** and sealed with a lid **216**. The fluidic channels **215** may be suitable for flowing gas and/or liquid within the body **201**. This can be used for cooling or other thermal control of the ESC **200**. In some embodiments, multiple zones of fluidic channels **215** may be used in order to provide more refined thermal control across an entire surface of the ESC **200**.

[0034] In an embodiment, a ceramic plate **205** is provided on the body **201**. The ceramic plate **205** may sit in a recess of the second surface **204**, or the ceramic plate **205** may be over the first surface **204**. The ceramic plate **205** may comprise any suitable ceramic material for ESC chucking, such as aluminum nitride, aluminum oxide, or the like. In an embodiment, an electrode **208** may be embedded in the ceramic plate **205**. The electrode **208** may be an electrically conductive material, such as copper or the like. The electrode **208** may be a conductive plate, a conductive mesh, or have any other conductive pattern that is distributed though the ceramic plate **205**. In the illustrated embodiment, the electrode **208** is set at a midpoint along a thickness direction of the ceramic plate **205**. Though, in other embodiments the electrode **208** may be closer to a top of the ceramic plate **205** or closer to a bottom of the ceramic plate **205**. When the electrode **208** is embedded within the ceramic plate **205**, as shown in FIG. **2**A, the ESC **200** may sometimes be referred to as an embedded ESC **200**.

[0035] In an embodiment, the electrode **208** may be electrically coupled to a DC input **230**. The DC input **230** may be an electrically conductive pin that passes through at least a thickness of the body **201**. The DC input **230** may pass through a portion of the ceramic plate **205** in order to directly contact the electrode **208**. In other embodiments, the DC input **230** may be electrically coupled to the electrode **208** by a via (not shown) that passes through at least a portion of the thickness of the ceramic plate **205**.

[0036] In an embodiment, the DC input **230** is electrically isolated from the sidewall of a hole **233** that passes through the body **201**. For example, a collar **225** may be provided between the DC input **230** and the sidewall of the hole **233**. The collar **225** may be an electrically insulating material, such as a polymer material, a dielectric material, and/or the like. In an embodiment, the collar **225** extends along an entire depth of the hole **233**. The presence of the collar **225** prevents the DC current from flowing into the body **201**. Accordingly, substantially all of the DC voltage/current is applied to the electrode **208** within the ceramic plate **205**. This reduces (or eliminates) leakage from the DC input **230**.
[0037] In an embodiment, the RF feed **220** is provided as a direct connection to the body **201**. The RF feed is spaced away from the DC input **230** and electrically isolated from the DC input **230**. As such, control of the RF bias applied to the ESC **200** is separate and independently controllable with respect to the DC bias applied to the electrode **208**. As such, electrical leakage of the ESC **200** is improved. This allows for lower chucking powers and improved energy efficiency.
[0038] Referring now to FIG. **2**B, a cross-sectional illustration of an ESC **200** that is coupled to a facility plate **222** is shown, in accordance with an embodiment. In an embodiment, the facility plate **222** is part of the larger pedestal structure on which the ESC **200** is attached. The facility plate **222** may be an interface structure that allows for the DC input **230** and the RF feed **220** to be passed to regions underlying the ESC **200** and out of the chamber (not shown).
[0039] In an embodiment, the facility plate **222** may comprise a metallic material, such as aluminum or the like. In an embodiment, a hole may pass through the facility plate **222** below the DC input **230**. In an embodiment, a connector assembly **227** may pass through the hole in the facility plate **222**. The connector assembly **227** may comprise an electrically insulating housing that surrounds a pin **228**. The pin **228** may be electrically coupled to the DC input **230**. In some embodiments, the pin **228** directly contacts the DC input **230**. In other embodiments, one or more intervening electrically conductive structures are provided between the pin **228** and the DC input **230**.
[0040] In an embodiment, the RF feed **220** passes through the facility plate **222** as well. In other embodiments, the RF feed **220** may have a first portion that contacts the facility plate **222**, and a second portion that contacts the body **201** of the ESC **200**. For example, an electrically conductive rod may contact the facility plate **222**, and a screw, a bolt, or the like may extend from the facility plate **222** to the body **201** of the ESC **200**. In this way, the rod is electrically coupled to the bolt through the facility plate **222**.
[0041] In the embodiment shown in FIGS. **2**A and **2**B, the DC input **230** is shown as a simple rod that passes through a hole **233** with a constant diameter. Such a construction may be suitable for some designs. However, in cases where the ESC is mounted to the facility plate with a blind assembly process, the design of the DC input may need to be modified. Examples of such embodiments are shown in FIGS. **3**A-**3**D.
[0042] Referring now to FIG. **3**A, a cross-sectional illustration of a portion of an ESC **300** is shown, in accordance with an embodiment. In an embodiment, the ESC **300** may comprise an electrically conductive body **301**. The conductive body **301** may be similar in material and dimension to any of the conductive bodies described in greater detail herein. In an embodiment, the conductive body **301** may have a first surface **303** (e.g., a bottom surface) and a second surface **304** (e.g., a top surface) that is opposite from the first surface **303**.
[0043] In an embodiment, the ESC **300** may comprise fluidic channels **315**. The fluidic channels **315** may be provided at a bottom of the ESC **300**, and sealed with a lid **316**. The fluidic channels **315** may be suitable for flowing gas and/or liquid within the body **301** to enable thermal control of the ESC **300**. In some embodiments, multiple zones of fluidic channels **315** may be used.
[0044] In an embodiment, a ceramic plate **305** is provided on the body **301**. The ceramic plate **305** may sit in a recess of the second surface **304**, or the ceramic plate **305** may be over the first surface **304**. The ceramic plate **305** may be similar to any of the ceramic plates described in greater detail

herein. In an embodiment, an electrode **308** may be embedded in the ceramic plate **305**. The electrode **308** may be similar to any of the electrodes described in greater detail herein.

[0045] In an embodiment, a hole **333** may be provided through the body **301** of the ESC **300**. The hole **333** may have a non-uniform diameter (or other dimension) through a depth of the hole **333**. For example, a diameter of the hole **333** near the first surface **303** may be greater than a diameter of the hole **333** near the second surface **304**. In the illustrated embodiment, the hole **333** has a stepped profile. Other embodiments may include a hole **333** with a sloping profile. The hole **333** may be configured to receive the DC input (not shown in FIG. **3**A).

[0046] In an embodiment, a second hole **313** may be provided into the body **301**. The depth of the second hole **313** may be shallower than a depth of the hole **333**. In an embodiment, the second hole **313** is configured to provide a coupling point for an RF feed (not shown in FIG. **3**A). For example, the second hole **313** may accommodate a bolt, screw, or rod that is carrying the RF bias to the ESC **300**.

[0047] Referring now to FIG. **3**B, a zoomed in illustration of the DC input assembly **340** inserted into the hole **333** of the ESC **300** is shown, in accordance with an embodiment. In an embodiment, the DC input assembly **340** may comprise an electrically conductive rod **330** that is electrically coupled to a DC input **332** by an electrically conductive spring **336**. The DC input **332** may be an electrically conductive pad, pin, or the like. The DC input **332** may be electrically coupled to the electrode **308** by a via **339** or the like.

[0048] In an embodiment, the rod **330**, the spring **336**, and the DC input **332** may be surrounded by an electrically insulating collar **334**, **335**, and **337** to prevent electrical shorting to the body **301**. The collar **334**, **335**, and **337** is shown as being three distinct parts. In other embodiments, a single electrically insulating component can be used as the collar, or a plurality of components can be coupled together to form the collar. As such, a direct and isolated path from the conductive rod **330** to the electrode **308** can be made in order to apply a DC bias to the electrode **308** for generating a chucking force on a substrate (not shown in FIG. **3**B).

[0049] In an embodiment, the design of the DC input assembly **340** may take into consideration assembly processes. For example, the ESC **300** is often attached to the underlying facility plate (not shown in FIG. **3**B) with a blind install. As such, the ability to provide alignment tolerance is beneficial in order to make assembly easier. In some instances, a portion of the collar **334** may include a recess **331** in order to accommodate a connector for the facility plate. The recess **331** may be a ring shaped recess along the bottom surface of the collar **334** that surrounds the rod **330**.

[0050] The presence of the spring **336** may also be beneficial for the assembly process. Particularly, the spring **336** provides a compressible member that allows for any variation in the placement of the ESC **300** in the Z-dimension to be accommodated. For example, if the ESC **300** is set too “high” the spring **336** expands to provide the proper electrical connection to the electrode **308**. Similarly, if the ESC **300** is set too “low” the spring **336** compresses in order reduce the height of the DC input line.

[0051] Referring now to FIG. **3**C, a cross-sectional illustration of the ESC **300** with the DC input assembly **340** inserted into the body **301** is shown, in accordance with an embodiment. The DC input assembly **340** in FIG. **3**C may be substantially similar to the DC input assembly **340** in FIG. **3**B. The ESC **300** in FIG. **3**C also illustrates the RF feed **320** inserted into the body **301**. As shown, the RF feed **320** is an electrically conductive rod, pin, screw, bolt, or the like. The RF feed **320** is configured to provide an RF bias to the body **301** of the ESC **300**. The DC input assembly **340** provides electrical isolation between the DC input and the RF feed **320** (i.e., due to the presence of the electrically insulating collar around the rod **330**, the spring **336**, and the DC input **332**).

[0052] Referring now to FIG. **3**D, a cross-sectional illustration of an ESC **300** that is coupled to a facility plate **322** is shown, in accordance with an embodiment. In an embodiment, the facility plate **322** is part of the larger pedestal structure on which the ESC **300** is attached. The facility plate **322** may be an interface structure that allows for the DC input assembly **340** and the RF feed **320** to be passed to regions underlying the ESC **300** and out of the chamber (not shown).

[0053] In an embodiment, the facility plate **322** may comprise a metallic material, such as aluminum or the like. In an embodiment, a hole may pass through the facility plate **322** below the DC input assembly **340**. In an embodiment, a connector assembly **327** may pass through the hole in the facility plate **322**. The connector assembly **327** may comprise an electrically insulating housing that surrounds a pin **328**. The pin **328** may be electrically coupled to the rod **330** of the DC input assembly **340**. In some embodiments, the pin **328** directly contacts the rod **330**. In other embodiments, one or more intervening electrically conductive structures are provided between the pin **328** and rod **330**.

[0054] In an embodiment, the connector assembly **327** may comprise a protrusion **326** at an upper edge of the connector assembly **327**. The protrusion **326** may be a ring-shaped protrusion that surrounds a perimeter of the pin **328**. The protrusion **326** may be sized to insert into the recess **331** of the collar **334**. The protrusion **326** and recess **331** interface may be used in order to align the ESC **300** to the facility plate **322** when a blind install or assembly is used.

[0055] In an embodiment, the RF feed **320** passes through the facility plate **322** as well. In other embodiments, the RF feed **320** may have a first portion that contacts the facility plate **322**, and a second portion that contacts the body **301** of the ESC **300**. For example, an electrically conductive rod may contact the facility plate **322**, and a screw, a bolt, or the like may extend from the facility plate **322** to the body **301** of the ESC **300**. In this way, the rod is electrically coupled to the bolt through the facility plate **322**.

[0056] Referring now to FIG. **4**, a cross-sectional illustration of a semiconductor processing tool **450** is shown, in accordance with an embodiment. In an embodiment, the semiconductor processing tool **450** may include a plasma processing tool, such as a plasma etching chamber, a plasma dicing chamber, a deposition chamber that uses plasma (e.g., plasma enhanced chemical vapor deposition (PECVD) or plasma enhanced atomic layer deposition (PEALD), etc.), a plasma treatment chamber, or the like.

[0057] In an embodiment, the tool **450** may comprise a chamber **451**. The chamber **451** may be suitable for supporting a vacuum pressure within the chamber **451** in order to support the generation of a plasma **457**. In an embodiment, the chamber **451** may comprise an ESC **400** that is supported over a pedestal **455**. The interior of the pedestal **455** (e.g., a facility plate, etc.) is omitted for simplicity. In an embodiment, the ESC **400** may be similar to any of the ESCs described in greater detail herein. For example, the ESC **400** may comprise a metallic body **401** with a ceramic plate **405** on a top surface of the metallic body **401**. A DC input **430** may pass through a hole in the body **401**, and an electrically insulating collar **425** may electrically isolate the DC input **430** from the body **401**. An RF feed **420** may be electrically coupled to the body **401**. In an embodiment, fluid feed lines **453** may be provided to the ESC **400** in order to provide gasses or liquids to the ESC **400** for thermal control purposes.

[0058] In an embodiment, a substrate **458** may be provided over the ESC **400**. The substrate **458** may be chucked to the ESC **400** through the use of the DC input **430** that is coupled to an electrode (not shown) in the ceramic plate **405**. The substrate **458** may be a wafer (e.g., a silicon wafer) or any other type of substrate used in a semiconductor processing environment.

[0059] In an embodiment, a showerhead **452** may be provided as a lid to the chamber **451** that is opposite from the ESC **400**. Processing gasses may be flown into the chamber **451** through the showerhead **452**. The showerhead **452** may be biased with RF or microwave frequencies in order to ignite the plasma **457** within the chamber **451**.

[0060] Referring now to FIG. **5**, a process flow diagram of process **560** for processing a substrate with a semiconductor tool that includes an ESC with a DC input that is electrically isolated from an RF input is shown, in accordance with an embodiment. In an embodiment, the process **560** begins with operation **561**, which comprises placing a substrate on an ESC, where the ESC has a DC input that is electrically isolated from an RF input. The ESC used in process **560** may be similar to any of the ESCs described in greater detail herein.

[0061] In an embodiment, the process **560** may continue with operation **562**, which comprises applying a chucking force to the substrate by activating the DC input. Since the DC input is electrically isolated from a remainder of the ESC, the chucking force needed to secure the substrate is reduced compared to existing ESC devices. This also allows for lower current leakage and provides a more efficient tool.

[0062] In an embodiment, the process **560** may continue with operation **563**, which comprises processing the substrate in a plasma environment during an application of an RF bias applied by the RF input. In an embodiment, the processing may include a plasma etching process, a plasma dicing process, or any other treatment process. In an embodiment, the chucking force to the substrate remains substantially constant during the processing of the substrate. Since there is substantially no leakage, the uniform chucking force can be maintained without changing a DC voltage of the DC input.

[0063] In an embodiment, the process **560** may continue with operation **564**, which comprises releasing the chucking force. In an embodiment, the chucking force may be released by reducing a voltage of the DC input. After the chucking force is released, the process **560** may continue with operation **565**. Operation **565** may comprise removing the substrate from the ESC. The substrate may be removed by a wafer handling robot or the like.

[0064] Referring now to FIG. **6**, a process flow diagram of a process **670** for assembling an ESC to a pedestal is shown, in accordance with an embodiment. In an embodiment, the process **670** may begin with operation **671**, which comprises aligning the ESC with an RF feed line and a DC input. The ESC may have a first hole that is aligned over the DC input and a second hole that is aligned over the RF feed line.

[0065] In an embodiment, the process **670** may continue with operation **672**, which comprises engaging the ESC with the DC input. In an embodiment, engaging the DC input with the ESC compresses a spring between the DC input and an electrode of the ESC. In an embodiment, the compression of the spring improves the electrical connection between the DC input and the electrode of the ESC.

[0066] In an embodiment, the process **670** may continue with operation **673**, which comprises directly contacting an electrically conductive body of the ESC with the RF feed line. In an embodiment, the RF feed line provides an RF bias to the ESC during processing. In an embodiment, the RF feed line and the DC input are electrically isolated from each other. For example, the DC input may be separated from the conductive body of the ESC by an electrically insulating collar, similar to those described in greater detail herein.

[0067] Referring now to FIG. **7**, a block diagram of an exemplary computer system **700** of a processing tool is illustrated in accordance with an embodiment. In an embodiment, computer system **700** is coupled to and controls processing in the processing tool. Computer system **700** may be connected (e.g., networked) to other machines in a Local Area Network (LAN), an intranet, an extranet, or the Internet. Computer system **700** may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. Computer system **700** may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated for computer system **700**, the term “machine” shall also be taken to include any collection of machines (e.g., computers) that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies described herein.

[0068] Computer system **700** may include a computer program product, or software **722**, having a non-transitory machine-readable medium having stored thereon instructions, which may be used to program computer system **700** (or other electronic devices) to perform a process according to embodiments. A machine-readable medium includes any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices, etc.), a machine (e.g., computer) readable transmission medium (electrical, optical, acoustical or other form of propagated signals (e.g., infrared signals, digital signals, etc.)), etc.

[0069] In an embodiment, computer system **700** includes a system processor **702**, a main memory **704** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **706** (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory **718** (e.g., a data storage device), which communicate with each other via a bus **730**.

[0070] System processor **702** represents one or more general-purpose processing devices such as a microsystem processor, central processing unit, or the like. More particularly, the system processor may be a complex instruction set computing (CISC) microsystem processor, reduced instruction set computing (RISC) microsystem processor, very long instruction word (VLIW) microsystem processor, a system processor implementing other instruction sets, or system processors implementing a combination of instruction sets. System processor **702** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal system processor (DSP), network system processor, or the like. System processor **702** is configured to execute the processing logic **726** for performing the operations described herein.

[0071] The computer system **700** may further include a system network interface device **708** for communicating with other devices or machines. The computer system **700** may also include a video display unit **710** (e.g., a liquid crystal display (LCD), a light emitting diode display (LED), or a cathode ray tube (CRT)), an alphanumeric input device **712** (e.g., a keyboard), a cursor control device **714** (e.g., a mouse), and a signal generation device **716** (e.g., a speaker).

[0072] The secondary memory **718** may include a machine-accessible storage medium **731** (or more specifically a computer-readable storage medium) on which is stored one or more sets of instructions (e.g., software **722**) embodying any one or more of the methodologies or functions described herein. The software **722** may also reside, completely or at least partially, within the main memory **704** and/or within the system processor **702** during execution thereof by the computer system **700**, the main memory **704** and the system processor **702** also constituting machine-readable storage media. The software **722** may further be transmitted or received over a network **761** via the system network interface device **708**. In an embodiment, the network interface device **708** may operate using RF coupling, optical coupling, acoustic coupling, or inductive coupling.

[0073] While the machine-accessible storage medium **731** is shown in an exemplary embodiment to be a single medium, the term "machine-readable storage medium" should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term "machine-readable storage medium" shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies. The term "machine-readable storage medium" shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0074] In the foregoing specification, specific exemplary embodiments have been described. It will be evident that various modifications may be made thereto without departing from the scope of the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. An apparatus, comprising: a body with a first surface and a second surface opposite from the first surface, wherein the body comprises an electrically conductive material; a hole into the first surface of the body; a ceramic plate on the second surface of the body, wherein an electrically conductive electrode is embedded in the ceramic plate; and a pin that is electrically conductive, wherein the pin is positioned in the hole and the pin is electrically isolated from the body, and wherein the pin is electrically coupled to the electrode embedded in the ceramic plate.
2. The apparatus of claim 1, wherein the pin is electrically coupled to the electrode by a spring within the hole.
3. The apparatus of claim 1, wherein the pin is electrically isolated from the body by a collar that is electrically insulating.
4. The apparatus of claim 3, wherein a bottom surface of the collar has a ring shaped recess.
5. The apparatus of claim 3, wherein the collar comprises two or more components.
6. The apparatus of claim 1, further comprising an input coupled to the body, wherein the input is configured to apply an RF bias to the body.
7. The apparatus of claim 1, wherein the pin is configured to apply a DC voltage to the electrode embedded in the ceramic plate.
8. The apparatus of claim 1, further comprising: a channel configured to flow a gas or fluid within the body.
9. The apparatus of claim 1, wherein the body has a diameter greater than 200 mm.
10. The apparatus of claim 1, wherein the hole has a non-uniform diameter through the body.
11. A method, comprising: placing a substrate on an electrostatic chuck (ESC), wherein the ESC has a DC input that is electrically isolated from an RF input; applying a chucking force to the substrate by activating the DC input; processing the substrate in a plasma environment during application of an RF bias applied by the RF input; releasing the chucking force; and removing the substrate from the ESC.
12. The method of claim 11, wherein processing the substrate comprises a plasma dicing operation, an etching operation, or a deposition operation.
13. The method of claim 11, wherein the DC input comprises a pin that is surrounded by an electrically insulating collar.
14. The method of claim 13, wherein the DC input further comprises a spring between the pin and an electrode embedded in the ESC.
15. The method of claim 11, wherein the chucking force to the substrate remains substantially constant during the processing of the substrate without changing a DC voltage of the DC input.
16. The method of claim 11, wherein the ESC comprises at least two backside gas zones.
17. The method of claim 11, wherein the RF input biases an electrically conductive body of the ESC.
18. A semiconductor processing tool, comprising: a chamber suitable for maintaining a vacuum environment within the chamber; and an electrostatic chuck (ESC) within the chamber, wherein the ESC comprises: a body that is electrically conductive; a ceramic plate on the body, wherein an electrically conductive electrode is embedded within the ceramic plate; a DC input electrically coupled to the electrode, and wherein the DC input is electrically isolated from the body; and an RF input electrically coupled to the body.
19. The semiconductor processing tool of claim 18, wherein the semiconductor processing tool is configured for plasma dicing operations, etching operations, or deposition operations.
20. The semiconductor processing tool of claim 18, wherein the DC input comprises a pin, and wherein the pin is electrically coupled to the electrode by at least a spring.